

# Alberto Mitroi

## Full Stack Software Engineer | C#.NET Developer | Azure, Angular, React, TypeScript, Python | Applied AI

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### SUMMARY

Full Stack Software Engineer with 4 years of hands-on C#.NET development experience and a strong Python automation background. Builds production-facing .NET 8 services, REST APIs, microservices, desktop automation tools and React/Angular applications. Experience spans distributed and event-driven systems, Microsoft Azure, Docker, CI/CD, unit and integration testing, workflow automation and applied AI across product teams, startup delivery and independent software development. Microsoft Certified: Azure Developer Associate (AZ-204).

### CORE SKILLS

**Languages:** C#, Python, TypeScript, JavaScript, SQL, VBA, HTML, CSS

**C# and .NET:** .NET 8, .NET Core, ASP.NET Core, Web API, REST APIs, Entity Framework Core, LINQ, async/await, Dependency Injection, MediatR, FluentValidation, xUnit

**Architecture:** Microservices, Domain-Driven Design, CQRS, Clean Architecture, Vertical Slice Architecture, Event-Driven Architecture, Event Sourcing, SOLID, OOP, Design Patterns

**Messaging and data:** Kafka, RabbitMQ, Azure Service Bus, PostgreSQL, SQL Server, Marten, Elasticsearch, Redis, Database Modeling, Migrations

**Frontend:** Angular, React, Next.js, TypeScript, JavaScript, HTML, CSS, Tailwind CSS, tRPC, Prisma ORM

**Azure and DevOps:** Microsoft Azure, Container Apps, App Service, Azure SQL, Blob Storage, Key Vault, Application Insights, Azure DevOps, GitHub Actions, CI/CD, Docker, Git

**Automation and scripting:** Python, CLI automation, process orchestration, Windows desktop development, GUI development, file processing, data parsing, report generation, logging

**Applied AI:** Azure OpenAI, RAG, Large Language Models, Semantic Search, Embeddings, Conversational AI, Voice AI, AI Agents, Prompt Engineering

**Practices:** Agile, Scrum, SDLC, Code Reviews, Unit Testing, Integration Testing, Debugging, Observability, Technical Documentation

### EXPERIENCE

#### Encora Inc. (part of Coforge) | Craiova, Romania (Hybrid)

.NET Full Stack Developer | Jul 2024 - Present | Full-time

- Engineer and support production .NET 8 services, REST APIs and Angular features for a distributed insurance platform serving the UK market.
- Delivered .NET 8 backend services and Angular features for a trading application using dependency injection, Clean Architecture, automated testing and code reviews.
- Design and implement asynchronous domain workflows with Kafka, RabbitMQ and Azure Service Bus across a microservices architecture.
- Model event-sourced workflows with CQRS, MediatR, Domain-Driven Design, Marten and PostgreSQL; build operational and search views in Elasticsearch.
- Own work through delivery and operations, including unit and integration testing, Azure CI/CD releases, incident analysis and reliability improvements.
- Work in an Agile/Scrum team across the full SDLC, from refinement and design through code review, testing and production support.

#### Oryntech | Craiova, Romania (Remote)

AI Solutions Engineer & Co-CTO | Mar 2026 - Present | Part-time

- Co-lead technical architecture and build full-stack AI product features and workflow automations for service businesses.
- Deliver React, Next.js and TypeScript interfaces integrated with REST APIs, webhooks, CRM and booking workflows.
- Implement voice and chat agent flows with LLM integration and prompt engineering for customer conversations, lead follow-up and multi-channel messaging.
- Harden AI workflows with context management, guardrails, human handoff, testing and reliability controls.

#### FORVIA HELLA | Craiova, Romania (Hybrid)

Software Automation Developer | Nov 2023 - Jul 2024

- Developed C#, .NET and Python automation tools and scripts that reduced repetitive manual engineering work.
- Built internal desktop GUI applications with C# and VBA, and automated data processing, file handling and report generation.
- Tested automotive control-system software and improved validation workflows in collaboration with engineering teams.

## Independent Developer | Craiova, Romania (Remote)

Software Developer - C# / Python | Sep 2022 - Nov 2023 | Part-time

- Developed a Windows desktop application in C#.NET for automating and orchestrating production-preparation workflows in a metalworking and CNC environment.
- Integrated Python scripts and CLI-based utilities for file processing, job validation, data parsing and report generation.
- Implemented process execution and orchestration, command-line argument management, stdout/stderr capture, asynchronous background tasks, logging, validation and error handling.
- Built a graphical interface that allowed users to configure, launch and monitor automation workflows without interacting directly with command-line tools.
- Structured reusable and maintainable components using OOP, SOLID principles, async/await, dependency injection and configuration management.

**Technologies:** C#, .NET, Python, Windows desktop development, CLI automation, process orchestration, async/await, OOP, SOLID, dependency injection, logging, Git

## EDUCATION

### Faculty of Automation, Computers and Electronics | University of Craiova

Master's degree, Artificial Intelligence and Applied Computing | Aug 2026 - Jul 2028

Bachelor of Engineering, Computer Science and Information Technology | Sept 2022 - Jul 2026

## CERTIFICATIONS

- Microsoft Certified: Azure Developer Associate (AZ-204) | Microsoft | Credential ID 42EFC399D66B1E5C | Mar 2026
- Introduction to Generative AI for Software Development | DeepLearning.AI | Credential ID G503GKSPZ3IO | Jul 2025
- Generative AI with Large Language Models | Amazon Web Services | Credential ID A3COZZP9ND79 | Jul 2025

## PROJECT EXPERIENCE

### EShopMicroservices

[github.com/AlbertoMitroi/EShopMicroservices](https://github.com/AlbertoMitroi/EShopMicroservices)

- Built a .NET 8 and C# 12 e-commerce reference architecture with ASP.NET Core, Entity Framework Core and REST APIs.
- Applied Domain-Driven Design, CQRS, Clean Architecture, Vertical Slice Architecture, MediatR, FluentValidation and automated testing.
- Implemented RabbitMQ, Kafka and Redis communication, Docker packaging, GitHub Actions CI/CD and Azure-oriented deployment patterns.

**Technologies:** .NET 8, C# 12, ASP.NET Core, Web API, Entity Framework Core, Microservices, DDD, CQRS, MediatR, FluentValidation, RabbitMQ, Kafka, Redis, Docker, GitHub Actions, Azure

### Oryntech.AI

[oryntech.ai](https://oryntech.ai) | Full-stack product and applied AI work

- Develop React, Next.js and TypeScript interfaces for AI-assisted service-business workflows.
- Build conversational and voice-agent features with LLM integration, context management, guardrails, human handoff and reliability controls.
- Integrate CRM, email, SMS, booking, analytics, APIs and webhooks into multi-channel automation workflows.

**Technologies:** React, Next.js, TypeScript, Tailwind CSS, Framer Motion, OpenAI, LLM Integration, Conversational AI, Voice AI, CRM, APIs, Webhooks, Workflow Automation

### TopZonal

Independent full-stack product | In development

- Build a mobile-first local-services PWA with React, Next.js, TypeScript, tRPC, Prisma ORM and a monorepo architecture.
- Design Azure-oriented components using Docker, Azure Container Apps, Azure SQL, Blob Storage and Azure Key Vault.
- Prototype semantic search, embeddings, vector retrieval and RAG-assisted discovery with Azure OpenAI.

**Technologies:** React, Next.js, TypeScript, T3 Stack, tRPC, Prisma ORM, PWA, Docker, Azure Container Apps, Azure SQL, Blob Storage, Key Vault, Azure OpenAI, RAG